



Kirk Tolleshaug

Senior Backend Software Engineer | C#/.NET | APIs, Distributed Systems & Architecture

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Summary

Senior backend software engineer with 12+ years building reliable, high-performance systems in C#/.NET: REST APIs, distributed and event-driven systems, relational databases, cloud-hosted applications, and legacy modernization.

Recent work centers on .NET backend services for web, mobile, external integrations, and high-volume event processing across systems serving tens of thousands of users and millions of devices, spanning API design and versioning, secure authentication, distributed processing, observability, and production reliability.

I've taken API response times from several seconds to sub-second, cut long-running processing from hours to minutes and minutes to seconds, modernized legacy .NET services, and built resilient event-driven workflows, while contributing architecture ownership, technical design, and mentoring.

Technical Skills

Languages & Frameworks: C#, .NET, ASP.NET, Go, Java, Angular, React

APIs & Distributed Systems: REST APIs, API Versioning, OpenAPI/Swagger, Azure API Management, Distributed Systems, Event-Driven Architecture, gRPC, RabbitMQ, Background Services, Multithreading, Parallel Processing

Reliability & Observability: Idempotency, Retries, Dead-Letter Queues, New Relic, Structured Logging, Metrics, Service Health Monitoring

Data & Messaging: Microsoft SQL Server, PostgreSQL, Entity Framework, Schema Migrations, Redis, Distributed Caching

Cloud & Delivery: Azure (App Services, SQL, Key Vault, Functions), AWS, Docker/Compose, Infrastructure as Code (Pulumi, Bicep), CI/CD, IIS

Security & Quality: OAuth/SSO, JWT, Roles and Claims, Service-to-Service Authentication, TDD (NUnit, xUnit), Integration/End-to-End/Performance Testing

Professional Experience

Digital Monitoring Products

Software Engineer

December 2022 - Present · Springfield, MO

Work across backend services, APIs, distributed processing, device integrations, delivery automation, and web applications.

- Design and optimize C#/.NET backend services and REST APIs for internal web and mobile applications and external integrations, supporting tens of thousands of users and handling more than 100M API requests each month.
- Design and version public REST APIs, introducing v2 routes and schemas while deprecating v1 endpoints; maintain OpenAPI/Swagger documentation and configure Azure API Management with JWT validation, subscription keys, rate limits, and request transformations.
- Implement authentication services and standardize JWT handling through internal libraries, establishing Azure Key Vault usage for secure application secret management.
- Diagnose performance bottlenecks across memory, request processing, concurrency, and workload distribution, reducing operations from hours to minutes, minutes to seconds, and selected API paths to sub-second response times.
- Build resilient RabbitMQ event and job workflows using dead-letter queues, duplicate-event protection, retry logic, pending-message validation, and idempotent API handling.
- Modernize legacy services from .NET Framework 4.x and .NET Core 3 to modern .NET, and containerize local dependencies with Docker Compose to improve CI/CD reliability and cross-team development compatibility.
- Use AI-assisted engineering tools across unfamiliar code paths, legacy refactoring, defect investigation, change review, log and trace analysis, dependency mapping, API and schema compatibility reviews, test-case generation, and technical documentation, with developer-owned validation through tests and code review.
- Design SQL queries, stored procedures, and indexes across SQL Server and PostgreSQL; manage PostgreSQL schema migrations, backward-compatible changes, data backfills, and production rollouts.

- Build observability with Serilog and New Relic, publishing metrics and dashboards for service performance and health.
- Develop React-based user interfaces and application workflows supporting backend services and operational needs.
- Design a Go-based security-camera integration using REST APIs and bidirectional gRPC; prototype mesh communication for relaying messages during connectivity loss.
- Provide technical leadership through architecture, design discussions, project decomposition, mentoring, and engineering/QA coordination.

Vizient, Inc

Senior Software Engineer

June 2014 - December 2022 · Cape Girardeau, MO

Progressed from software engineering intern to senior software engineer, taking on architecture ownership, technical leadership, and end-to-end delivery across identity, reporting, modernization, and Azure-hosted applications.

- Owned technical design and end-to-end delivery across multiple initiatives as primary technical owner for requirements, system design, implementation, deployment, and support; decomposed complex work, mentored engineers, and coordinated with QA.
- Built cloud spend and usage reporting that identified unnecessary services and optimization opportunities, contributing to approximately \$30M in realized annual infrastructure savings.
- Developed C#/.NET applications and backend services across multiple generations of .NET, with Angular front ends and SQL Server data stores; designed Entity Framework migrations for greenfield Azure applications and contributed to Flyway-based legacy migrations.
- Applied TDD with NUnit and xUnit across new features, bug fixes, and modernization, using unit, integration, end-to-end, and performance testing in local and CI validation; built Docker-based test environments for dependent services.
- Improved backend and API performance, reducing select response times from several seconds to sub-second performance.
- Built Azure-hosted applications using App Services, Azure SQL, Key Vault, and Functions, with reusable infrastructure components and CI/CD automation (Pulumi, early Bicep) to standardize provisioning and deployment.
- Designed internal authentication services and OAuth/SSO integrations managing JWT lifecycle, roles and claims, service-to-service authentication, user management, Key Vault-backed secret rotation, API documentation, versioning, and backward compatibility.
- Designed and delivered a reporting system combining multiple domains and schemas, performing extensive customer-specific transformations and processing datasets containing millions of records.

Southeast Missouri State University

Teaching Assistant

January 2015 - December 2015 · Cape Girardeau, MO

- Assisted instruction for Computer Science I-III, C and Unix, and Logic Fundamentals; tutored students on programming and debugging, and graded coursework.

WebParaprofessional

January 2014 - May 2015 · Cape Girardeau, MO

- Built ASP.NET/C# web applications on SQL Server, Entity Framework, and IIS for university departments, including an end-to-end sign-in and usage-tracking system for the university writing lab; supported Moodle for faculty and staff.

Education

Southeast Missouri State University

Bachelor of Science, General Studies

2015 · Cape Girardeau, MO

Certification

Microsoft Certified: Azure Fundamentals (AZ-900)  - Microsoft · Issued December 2021